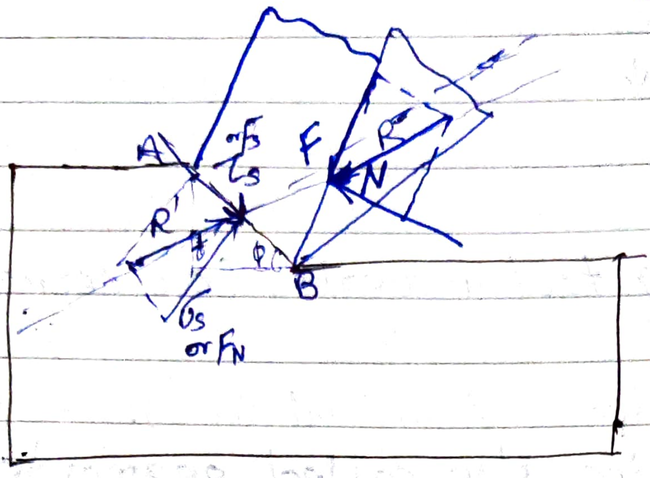
# 2. Force and Power in Metal Cutting

[for orthogonal metal cutting]




F = friction force
N = Normal force

$F = \mu N$

Length of shear plane (AB)

$t = AB \sin\phi$

or $AB = \frac{t}{\sin\phi}$

Area of shear plane $(A_s) = \frac{bt}{\sin\phi}$

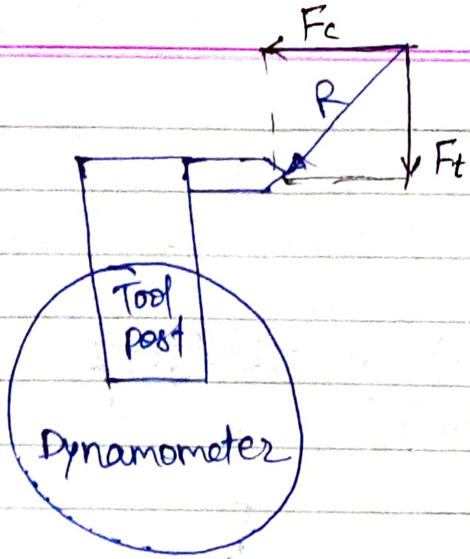
Shear ~~stress~~ Force on shear plane

$$F_s = \tau_s \times \frac{bt}{\sin\phi}$$

Normal force on shear plane

$$F_N = \sigma_s \times \frac{bt}{\sin\phi}$$

Chip is moving with constant velocity ($V_c$) therefore net resultant force on the chip must be zero. Therefore R and R' must be equal and opposite.

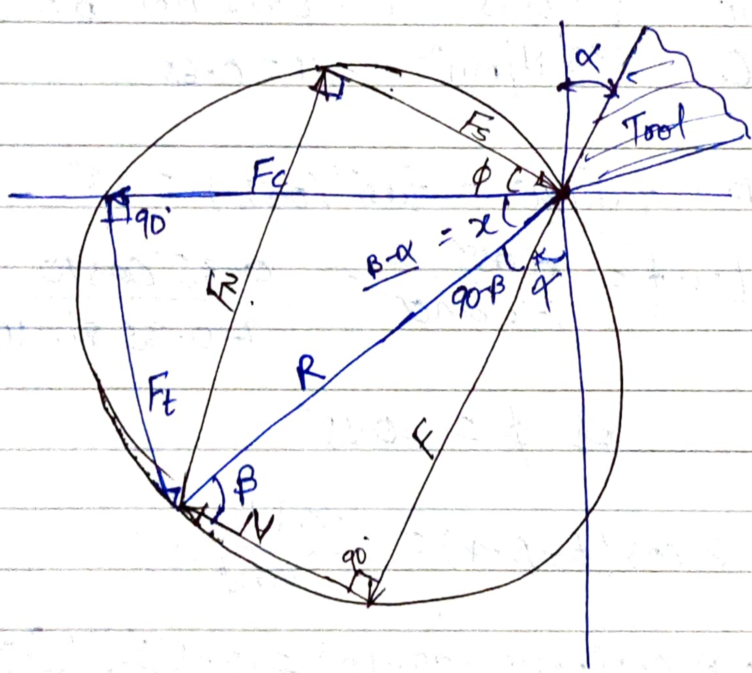



Cutting force or Horizontal force ($F_c$)

Thrust force or vertical force ($F_t$).

Merchant force circle diagram (MCD).

Resultant R as a diameter




$$\tan\beta = \frac{F}{N} = \mu$$

or $\beta = \tan^{-1}(\mu)$ = Friction angle.

$$x + 90 - \beta + \alpha = 90$$

or $\underline{x = \beta - \alpha}$

$F = R\sin\beta$

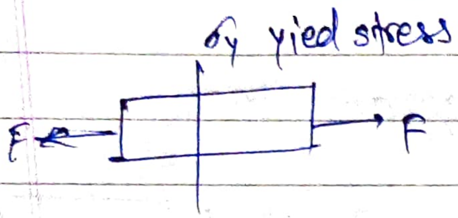
$N = R\cos\beta$

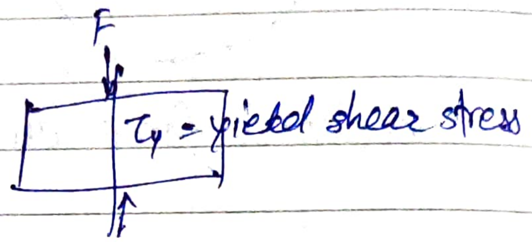
$F_c = R\cos(\beta-\alpha)$

$F_t = R\sin(\beta-\alpha)$

$F_s = R\cos(\phi+\beta-\alpha) = \tau_s \cdot \dfrac{bt}{\sin\phi}$

$F_N = R\sin(\phi+\beta-\alpha)$




Ques $\alpha = 10^\circ$ $\quad r = 0.35$ $\quad t = 0.51$ mm $\quad b = 3$ mm

$\tau_s = 285$ N/mm$^2$ $\quad \mu = 0.65$

$\beta = \tan^{-1}(\mu) = \tan^{-1}(0.65) = 33^\circ$

$\tan\phi = \dfrac{r\cos\alpha}{1-r\sin\alpha} = \dfrac{0.35\cos 10}{1-0.35\sin 10}$

or $\phi = 20.15^\circ$

$F_s = R\cos(\phi+\beta-\alpha) = \dfrac{\tau_s bt}{\sin\phi}$

$= R\cos(20.15+33-10) = \dfrac{285 \frac{N}{mm^2} \times 3\,mm \times 0.51\,mm}{\sin(20.15)}$

$F_s = 1265.8$ N

or $R = \dfrac{1265.8}{\cos(20.15+33-10)} = 1735.3$ N

(i) cutting force $F_c = R\cos(\beta-\alpha)$

$= 1735.3\cos(33-10)$

$= 1608$ N

[ $F = \mu N$ by Coulomb's Law for sliding friction

(ii) Radial force [Only in turning operation]
$= 0$

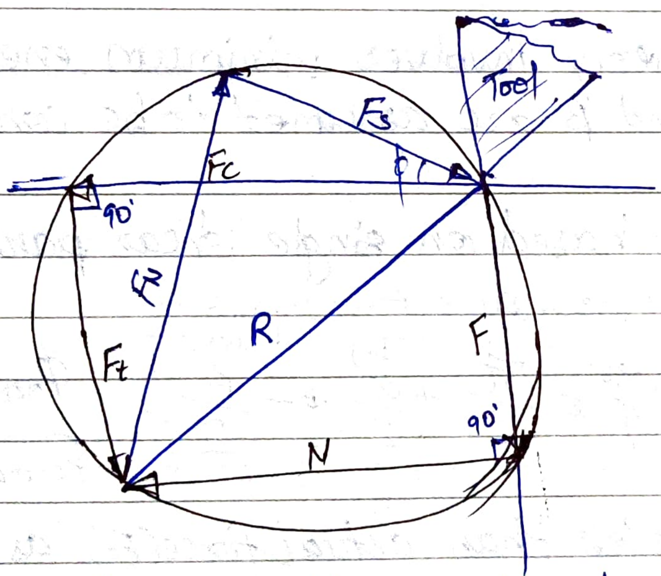
(iii) Normal force $(N) = R\cos\beta$
$= 1735.3 \cos 33 = 1456\ N$

(iv) shear force $(F_s) = 1265.8\ N$

Limitations of use of MCD

1. MCD is valid only for orthogonal cutting.
2. By the ratio F/N, the MCD gives apparent (not actual) co-efficient of friction.

If Rake angle $\alpha = 0$.




Merchant circle gives the rectangle

$$\boxed{\begin{aligned} F_c &= N \\ F_t &= F \end{aligned}}$$

Teacher's Signature

If $\alpha = 0$ & $\beta = 45°$

In merchant circle gives square

then

$$\boxed{F_c = F_t = F = N}$$

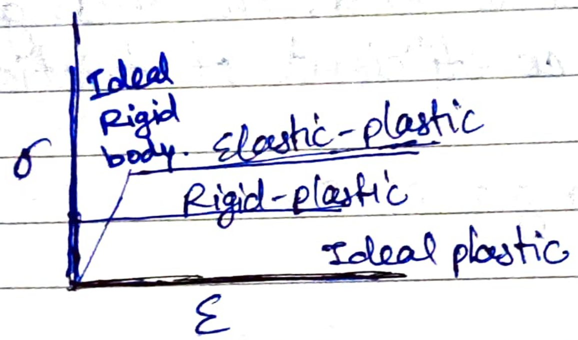
$\alpha = 0, \beta = 45°$ i.e. $\mu = 1$

Merchant Analysis - or Merchant Theory.

Different from merchant cycle.

Assumption.

- The work material behaves like an ideal plastic.




- The theory involves minimum energy principal.
- $\tau_s$ and $\beta$ are assumed to be constant, independent of $\phi$.
- It is based on single shear plane theory.

From Merchant Analysis -

$$\boxed{\underset{\text{Deg.}}{\phi} = \underset{\text{rad.}}{\frac{\pi}{4}} + \underset{\text{Deg.}}{\frac{\alpha}{2}} - \underset{\text{Deg.}}{\frac{\beta}{2}}}$$

$\phi_{merchant\ theory} \neq \phi_{theoreti.}$

$F_{c\,merch} < F_{c\,theoretical}.$

$F_c$ is less than actual force($F_c$) as in theoretical

$$\phi = 45 + \frac{\alpha}{2} - \frac{\beta}{2}$$

$$2\phi = \frac{\pi}{2} + \frac{\alpha}{2} - \beta$$

$$\boxed{2\phi + \beta - \alpha = \frac{\pi}{2}}$$

Ans $\alpha = 10$ $\tan\phi = \frac{r\cos\alpha}{1 - r\sin\alpha}$ $r = \frac{t}{t_c} = \frac{0.2}{t_c}$ | ESE

Applying Merchant theory.

$$\phi = 45 + \frac{\alpha}{2} - \beta/2$$

$$\beta = \tan^{-1}(\mu) = \tan^{-1}(0.5) = 26.56.$$

(i) shear angle $\phi = 45 + \frac{10}{2} - \frac{26.56}{2} = 36.7°$

(ii) $F_s = \frac{\tau_s bt}{\sin\phi} = \frac{400 N/mm^2 \times 2 mm \times 0.2 mm}{\sin 36.7}$

$$= 267.5 N$$

$$F_s = R\cos(\phi + \beta - \alpha)$$

or $267.5 = R\cos(36.7 + 26.56 - 10)$

or $R = 447.5 N$

Cutting force $F_c = R\,\cancel{\cos(\phi - \alpha)}$ $R\cos(\beta - \alpha)$

$= 447.5 \cos(36.7 - 10) = 429 N$

Thrust forse $F_t = R\sin(\beta - \alpha)$

$= 447.5\,\cancel{\cos}(36.7 - 10) = 127 N$

Modified Merchant Theory:

$$\tau_s = \tau_{s0} + K\sigma_s = F_s/A_s + K\frac{F_N}{A_s}$$

where, $\sigma_s$ is normal stress on shear plane $\left[\sigma_s = \frac{F_N}{A_s}\right]$

and then $\boxed{2\phi + \beta - \alpha = \cot^{-1}(K)}$

Teacher's Signature

## Theory of Lee and Shaffer:

- They applied the theory of plasticity for an ideal-rigid-plastic body.
- They also assumed that deformation occured on a thin-shear plane.

And derive.

$$\boxed{\phi = \frac{\pi}{4} + \alpha - \beta}$$

~~$F_c > F_{actual}$~~.

Other relations.

- By stabber

$$\phi = \frac{\pi}{4} + \frac{\alpha}{2} - \beta$$

The force relations - (VIMP)

$F = F_c \sin\alpha + F_t \cos\alpha$

$N = F_c \cos\alpha - F_t \sin\alpha$

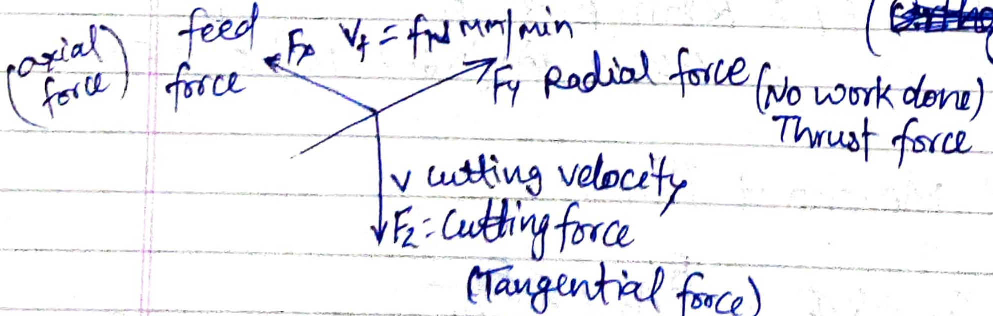
when $F_c$ & $F_t$ are known.

$F_N = F_c \sin\phi + F_t \cos\phi$

$F_s = F_c \cos\phi - F_t \sin\phi$

and $\mu = \frac{F}{N} = \tan\beta$

## Turning Operation:- (Oblique Cutting) (~~Orthogonal Cutting~~).




feed velocity $V_f = fN$ mm/min

$F_z \times V =$ Cutting power (about 99.6%)

$F_x \times V_f =$ feed power (0.4 Negligible)

Very Low

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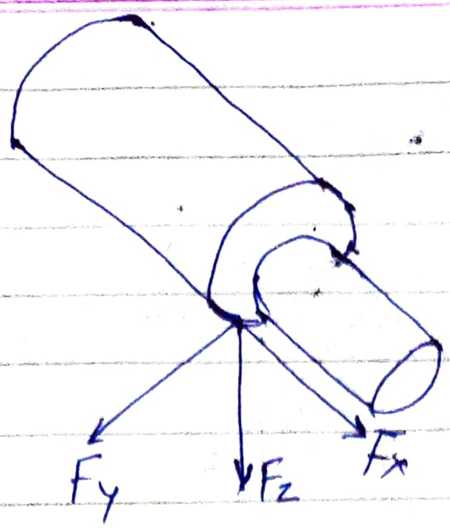


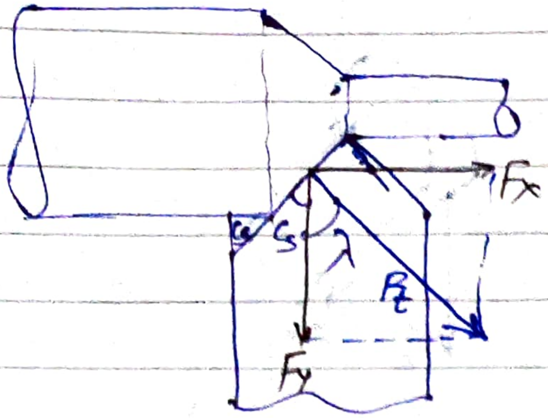

- $F_c$ or $F_z$ or Tangential ~~force~~ component or primary cutting force acting in the dir^n of the cutting velocity, largest force and accounts for 99% of the power required by the process.
- $F_f$ or $F_x$ or axial component or feed force acting in the dir^n of the tool feed. This force is about 50% of $F_c$ but accounts for only a small % of the power required because feed rates are usually small compare to cutting speeds.
- $F_r$ or $F_y$ or radial force or thrust force in turning acting ⊥ to the machined surface. This force is about 50% of $F_f$ i.e 25% of $F_c$ and contributes very little to power requirements because velocity in the radial dir^n is negligible

Conversion Formula:
We have to convert turning (3D) to orthogonal cutting (2D).

$$\boxed{F_c = F_z}$$

$$\boxed{F_t = F_{xy} = \frac{F_x}{\sin\lambda} = \frac{F_y}{\cos\lambda} = \sqrt{F_x^2 + F_y^2}}$$

Proove:



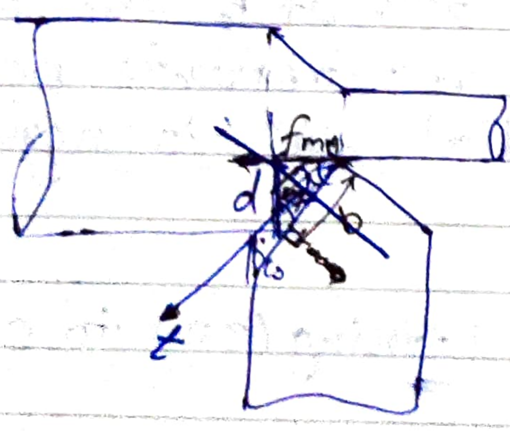

$$F_y = F_t \cos\lambda \quad \text{or} \quad F_t = \frac{F_y}{\cos\lambda}$$

$$F_x = F_t \sin\lambda \quad \text{or} \quad F_t = \frac{F_x}{\sin\lambda}$$

$$F_t = \sqrt{F_x^2 + F_y^2}$$

$$F_t = \frac{F_x}{\sin\lambda} = \frac{F_y}{\cos\lambda} = \sqrt{F_x^2 + F_y^2}$$

Geometry Conversion:
(limited ~~only~~ to ~~in~~ Orthogonal plane only).




feed = ~~f rev/mr~~
f rev/mm

d = depth of cut
b = width of cut

$d = b \sin\lambda$

$\boxed{b = d/\sin\lambda}$ — ①

$\boxed{t = f \sin\lambda}$ — ②

$\boxed{bt = fd}$

Q. $D = 160$ MM Diameter

Tool geometry $0, \underset{\alpha}{0}, 10, 8, 15, \underset{\lambda}{75}, 0 (mm)$

$F_z = F_c = 1200\,N$

$F_x = F_f = 800\,N$

$$F_t = \frac{F_x}{\sin\lambda} = \frac{800}{\sin 75} = 828\,N$$

$$b = \frac{d}{\sin\lambda} = \frac{4.0}{\sin 75} = 4.14\,mm$$

$$t = f\sin\lambda = 0.32 \sin\lambda = 0.3091\,mm.$$

(i) $F = F_c \sin\alpha + F_t \cos\alpha$

$= 1200 \sin 0 + 828 \cos 0$

$= 828\,N$

$N = F_c \cos\alpha - F_t \sin\alpha.$

$= 1200 \cos 0 - 828 \sin 0$

$= 1200\,N$

$$\mu = \frac{F}{N} = \frac{828}{1200} = 0.69$$

(ii) $F_s = F_c \cos\phi - F_t \sin\phi.$ Yield shear strength=?

$$r = \frac{t}{t_c} = \frac{0.3091}{0.8} = 0.386$$

$\alpha = 0$

$\tan\phi = r$, $\phi = \tan^{-1}(0.386) = 21.125°$

$F_s = 1200 \cos 21.125° - 828 \sin 21.125$

$= 821\,N$

$$\tau_s = \frac{F_s}{A_s} = \frac{F_s}{bt/\sin\phi} = \frac{F_s \sin\phi}{bt}$$

$$= \frac{821 \sin 21.125}{4.14 \times 0.3091} = 231 \frac{N}{MM^2}$$

(iii) Cutting power.

$$P = F_c \times V \;\; \text{(in m/s)}$$

$$= F_c \times \frac{\pi D N}{60}$$

$$= 1200 \times \frac{\pi \times 0.160 \times 400}{60}$$

$$= 4021 \text{ W}$$

$$= 4.021 \text{ kW} \quad \text{Ans.}$$

Ans. $D = 160$ mm.

$d = 2.5$ mm $\quad N = 315$ rpm.

$f = 0.16$ mm/rev.

Tool geometry $0, \underset{\alpha}{10}, 8, 9, 15, \underset{\lambda}{75}, 0$ (mm)

$F_c = 500$ N

$F_t = \frac{F_x}{\sin\lambda} = \frac{200\text{ N}}{\sin 75} = 207.05$ N

$b = \frac{d}{\sin\lambda} = \frac{2.5}{\sin 75} = 2.58$ mm

$t = f\sin\lambda = 0.16\sin 75 = 0.15$ mm

$F = F_c\sin\alpha + F_t\cos\alpha$

$= 500\sin 10 + 207.05\cos 10 = 290.67$ N

$N = F_c\cos\alpha - F_t\sin\alpha$

$= 500\cos 10 - 207.05\sin\alpha = 456.45$ N

$F_s = F_c\cos\phi - F_t\sin\phi$

$\cancel{\tan\phi} = r = \frac{t}{t_c} = \frac{0.15}{0.48} = 0.312$

$\tan\phi = \frac{r\cos\alpha}{1 - r\sin\alpha} = \frac{0.312\cos 10}{1 - 0.312\sin 10} =$

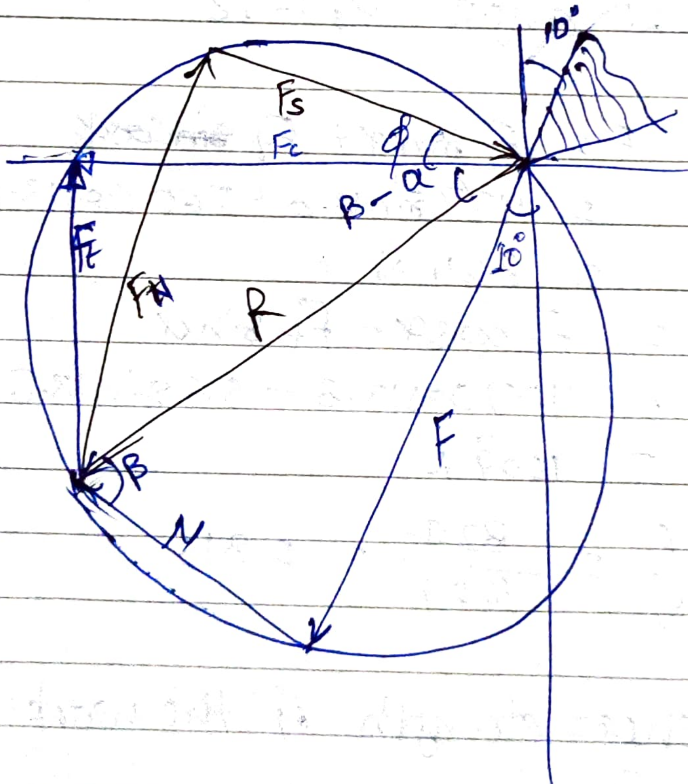
$\phi = \cancel{\tan^{-1}(}\ 18.\cancel{47}5$

$F_s = 500\cos 18.\cancel{47}5\ \cancel{\phi} - 207.05 \sin 18.\cancel{47}5$
$= \cancel{440.27\,N}\ 408.77\,N$

$F_N = F_c \sin\phi + F_t \cos\phi$
$= 500 \sin 18.\cancel{47}5 + 207.05 \cos 18.\cancel{47}5$
$= \cancel{351.37\,N}.\ 354.64\,N$

$\tan\beta = \frac{F}{N} = \frac{290.67}{456.45}$

$\beta = 32.48°$




$F = R\sin\beta \Rightarrow R = \frac{F}{\sin\beta} = \frac{290.67}{\sin 32.48}$

$= 541.28\,N$

$\lambda = 60^\circ$ $\quad \lambda = 90 - C_s$

Ques $\overset{\alpha_b}{7}, \overset{\alpha_s}{}, 6, 6, 8, \overset{C_s}{30}, 1$ (mm) ASA

$t_0 = 2.0$ mm $\quad b = 2.5$ mm.

$F_c = 1177$ N $\quad F_t = 560$ N

$\tan\alpha = \tan\alpha_s \sin\lambda + \tan\alpha_b \cos\lambda$

$\tan\alpha = \tan\alpha_s \sin 60 + \tan 7 \cos 60$ —— ①

$\tan\alpha_b = \tan\alpha \cos\lambda + \sin\lambda \tan i \quad i{=}0$

$\tan 7 = \tan\alpha \cos 60$

$\alpha = 13.8^\circ$

from eqⁿ ①

$\tan 13.8 = \tan\alpha_s \sin 60 + \tan 7 \cos 60$

or $\quad \alpha_s = 12^\circ$

(i) side rake angle $\alpha_s = 12^\circ$

(ii) $\mu = ?$ $\quad F = F_c \sin\alpha + F_t \cancel{\sin}\cos\alpha$

$= 1177 \sin 13.8 + 560 \cos 13.8$

$= 824$ N

$N = F_c \cos\alpha - F_t \sin\alpha$

$= 1177 \cos 13.8 - 56 \sin 13.8$

$= 1009$ N

$\mu = \frac{F}{N} = \frac{824}{1009} = 0.816$

(iii) Dynamic shear strength of the work material

$\tau_s = \frac{F_s}{A_s}$

$F_s = F_c \cos\phi - F_t \sin\phi$

Applying merchant theory.

$$\phi = 45 + \alpha/2 - \beta/2$$

$$\beta = \tan^{-1}(\mu) = \tan^{-1}(0.816) = 39.2^\circ$$

$$\phi = 45 + \frac{13.8}{2} - \frac{39.2}{2} = 32.3^\circ$$

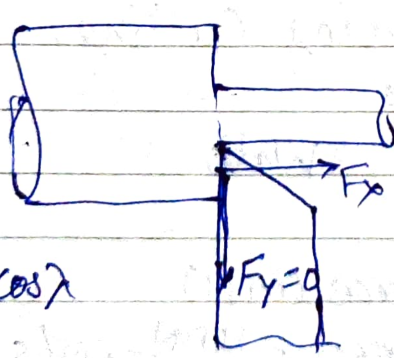
$$F_s = 1177 \cos 32.3^\circ - 560 \sin 32.3$$
$$= 696\,N$$

$$\tau_s = \frac{F_s}{A_s} = \frac{F_s}{bt/\sin\phi} = \frac{F_s \sin\phi}{bt}$$

$$= \frac{696\,N \sin 32.3}{2 \times 2.5\ mm.mm} = 74\ N/mm^2$$

Orthogonal Turning –

$$\boxed{\lambda = 90^\circ \text{ i.e. } C_s = 0}$$




$F_y = 0$

$F_x$ & $F_z$ exists.

$F_y = F_t \cos\lambda$
$= 0$

$$F_c = F_z$$
$$F_t = \frac{F_x}{\sin 90} = F_x$$
$$b = d$$
$$t = f$$

# Metal Removal Rate (MRR):

Metal Removal Rate (MRR) $= A_c \cdot V$ $\frac{mm^3}{min}$
$= btV$
$= fdV$

Where
$A_c$ = Cross-section area of uncut chip ($mm^2$)
$V$ = Cutting speed $= \pi DN$, mm/min

Ques: $V = 50$ m/min $f = 0.8$ mm/rev $d = 1.5$ mm
$MRR = fdV = 0.8 \times 1.5 \times 50 \times 1000$ $mm^3/min$
$= \cancel{60 \times 1000\ mm^3/min}$
$=$

Ques. $D = 200$ m $f = 0.25$ mm/rev $d = 4$ mm
$N = 160$ rpm
$$MRR = \frac{0.25 \times 4 \times \cancel{2000}\ \pi \times 200 \times 160}{60}\ mm^3/sec$$
$= 1675.5\ mm^3/sec$

## Power consumed During Cutting:

$$\boxed{P = F_c \times V} \quad \text{Watt}$$
($F_c$ in N, $V$ in m/s)

where
$F_c$ = Cutting force (in N)
$V$ = Cutting speed $= \frac{\pi DN}{60}$, m/s

Total Power = Shearing Power + Friction Power
$$\boxed{F_c \times V = F_s \times V_s + F \times V_c}$$
$F_s \times V_s$ → 60 – 75%
$F \times V_c$ → 30 – 35%

% of Frictional Power $= \frac{F \times V_c}{F_c \times V} \times 100\%$

% of Shearing Power $= \frac{F_s \times V_s}{F_c \times V} \times 100\%$

Specific Energy Consuption (J/mm³):

Conventional Machine (Lathe, Drill, Milling) → 2 - 3 J/mm³
Grinding → 50 J/mm³

Electrochemical Machine (ECM) → 500 J/mm³

Specific Energy Consuption $\boxed{e = \frac{\text{Power (J/s)}}{\text{MRR (mm}^3\text{/s)}}}$

$\boxed{e = \frac{F_c \times V \leftarrow m/s}{bt\,V \leftarrow mm/s}}$

$\boxed{e = \frac{F_c}{1000\,bt}} = \frac{F_c}{1000\,fd}$

Ans $e = 2.0\ J/mm^3$

$V_c = 120\ m/min \quad f = 0.2\ mm/rev \quad d = 2\ mm$

$2 = \frac{F_c}{1000 \times 2 \times 0.2} \Rightarrow F_c = 800\ N$

Ans. $D_o = 200\ m \quad D_i = 80\ m \quad f = 0.1\ mm/rev \quad d = 1\ mm.$

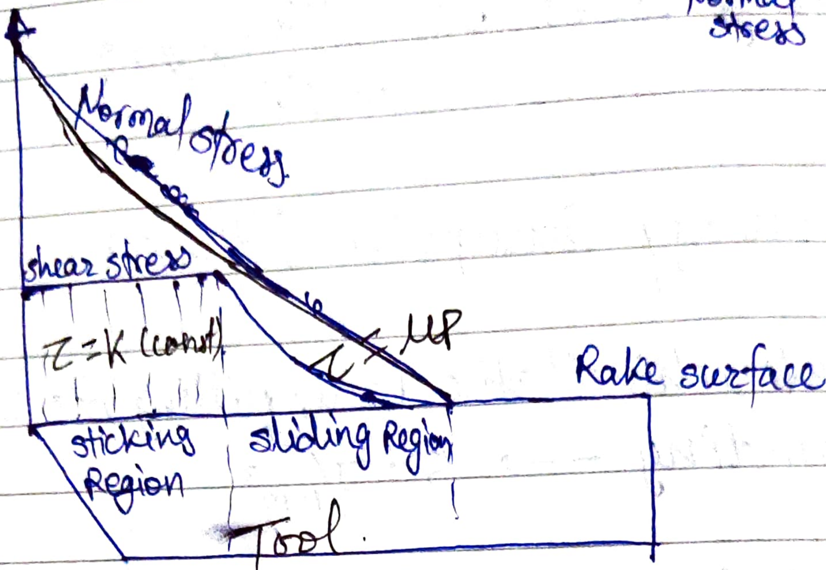
$V_c = 90\ m/min \quad F_c = 200\ N$

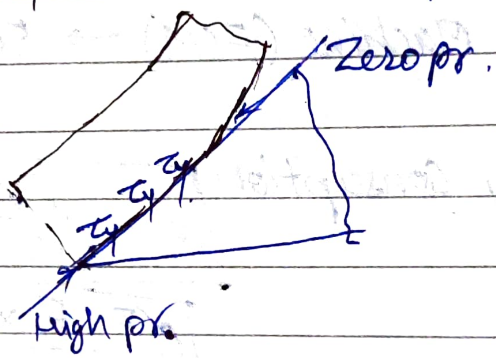
$e = \frac{F_c}{1000\,fd} = \frac{200}{1000 \times 0.1 \times 1} = 2\ J/mm^3$

# Friction in Metal Cutting:-

$\boxed{\sigma = -p}$

Normal stress — pressure




$\cancel{F = \mu N}$

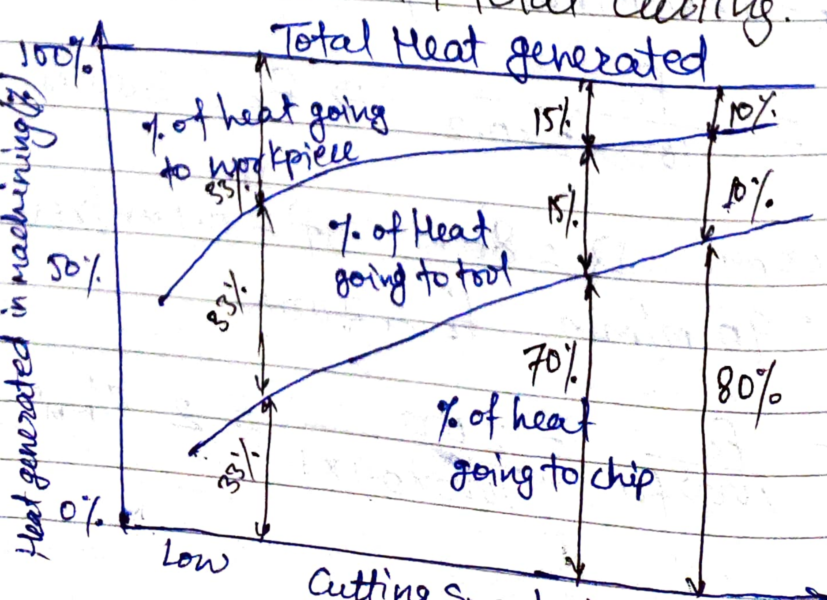
$F =$

$F = \mu N$

$F/A = \mu N/A$

$\boxed{\tau = \mu p}$ for sliding friction

## Heat Distribution in Metal Cutting:



Determination of Cutting heat & temp. Experimentally.

Heat

- Calorimetric method

Temperature.

- Decolouring agent
- Tool-work thermocouple
- Moving thermocouple technique
- Embedded thermocouple technique
- Using compound tool
- Indirectly from hardness and structral transformation
- Photo-cell technique
- Infraray detection method.

Dynamometer:

- Dynamometers are used for measuring Cutting forces.
- For ~~ot~~ orthogonal cutting use 2D dynamometer
- For oblique Cutting use 3D dynamometer.

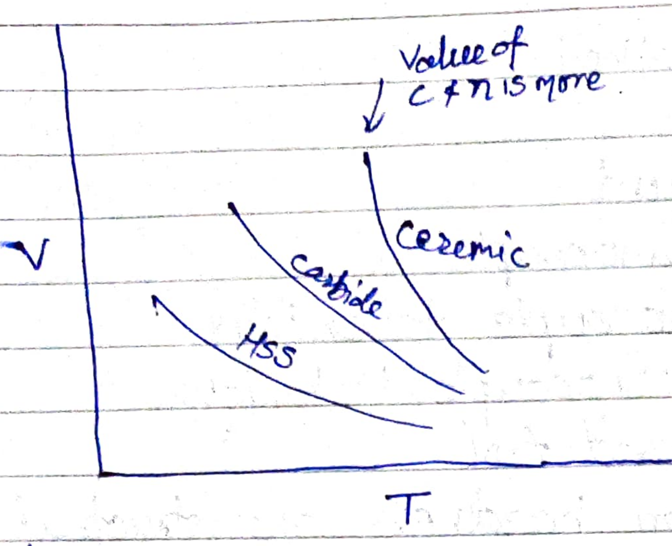
Types of Dynamometer.

- Strain gauge type
- Piezoelectric type

~~Gate-2008(PI)~~.

Carbon steel tools 250°C
Tungosten carbide tools 800°C
High speed steel tools 600°C.
Ceremic tools. 1200°C

# Tool life curve:-

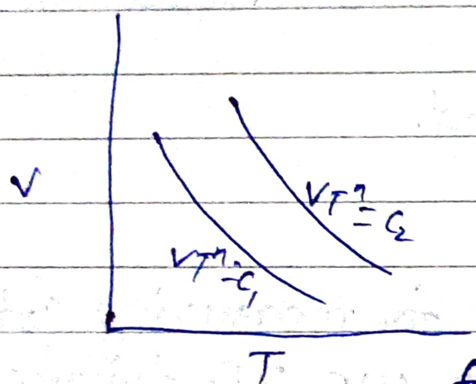


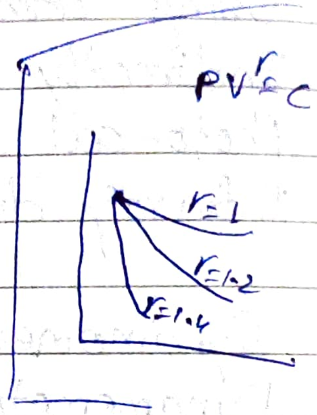

Taylor eqn

$$VT^n = C$$

$$yx^n = c$$

$$\text{or } y = \frac{c}{x^n}$$



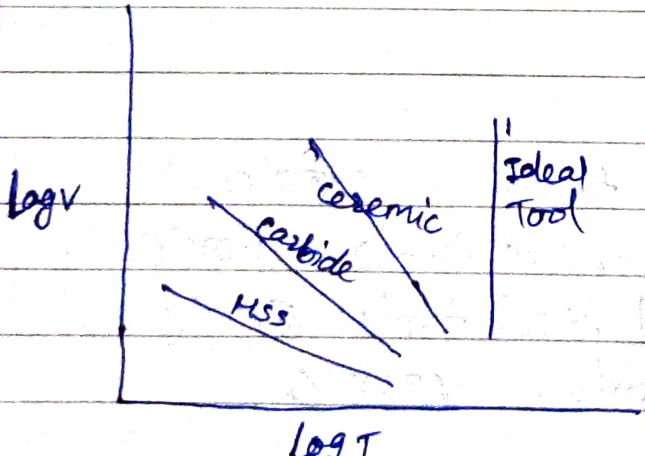




for same material (parallel)

$$VT^n = C$$

$$VT^n = C_1 = 100$$

$$VT^n = C_2 = 200$$

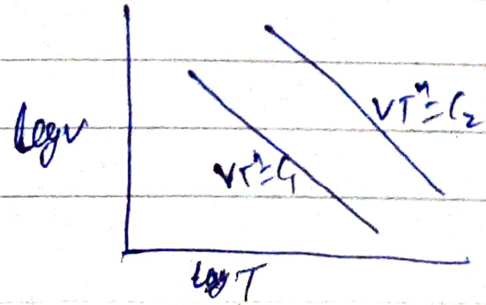



$$VT^n = C$$

$$\log V + n \log T = \log C$$

$$y + nx = K$$

$$y = -nx + K$$




for same material (∥)

$$C_2 > C_1$$

Gate 2010

Ans for Tool A $n = 0.45$ $K = 90$
for Tool B $n = 0.3$ $K = 60$

$$x T_A^{0.45} = 90 \Rightarrow T_A = \left(\frac{90}{x}\right)^{\frac{1}{0.45}}$$

$$x T_B^{0.3} = 60 \Rightarrow T_B = \left(\frac{60}{x}\right)^{\frac{1}{0.3}}$$

$$T_A > T_B \qquad \left(\frac{90}{x}\right)^{\frac{1}{0.45}} > \left(\frac{60}{x}\right)^{\frac{1}{0.3}}$$

or $$\left(\frac{90}{x}\right)^{\frac{1}{0.45}} = \left(\frac{60}{x}\right)^{\frac{1}{0.3}}$$

$$x = 26.67 \text{ m/min.}$$

Gate 2013

Carbide Tool $VT^{1.6} = 3000$
HSS Tool $VT^{0.6} = 200$

$$\left(\frac{3000}{V}\right)^{\frac{1}{1.6}} > \left(\frac{200}{V}\right)^{\frac{1}{0.6}}$$

$$\frac{1}{1.6} \log\left(\frac{3000}{V}\right) = \frac{1}{0.6} \log\left(\frac{200}{V}\right)$$

$$0.625 \log\left(\frac{3000}{V}\right) = 1.66 \log\left(\frac{200}{V}\right)$$

$$-3.7912 = \cancel{1.66\log}\ 0.625\cancel{\log V}\ \cancel{1.66}\log V$$
$$= \cancel{1.035 \log V}$$

$$\log V = 3.6629$$

$$V = \cancel{38.9}\ 39.4 \text{ m/min}$$

~~38.94 m/min~~

Que Cutting speed, m/min : 49.74, 49.23, 48.67, 45.76, 42.58 ($V_1$, $V_2$, $V_3$, $V_4$, $V_5$)
Tool life, min : 2.94, 3.90, ~~3.90~~ 4.77, 987, 28.27 ($T_1$, $T_2$, $T_3$, $T_4$, $T_5$)

$C = ?$

$VT^n = C$ $\quad \log V + n \log T = \log C$
$x + ny = K$

~~Regr~~ Linear Regration method is used

| | V | T | $x = \log V$ | $y = \log T$ | $xy$ | $x^2$ |
|---|---|---|---|---|---|---|
| 1 | 49.74 | $T_1 = 2.94$ | $x_1 = 1.6967$ | $y_1 = 0.4683$ | ~~1.8301~~ | 15.272 |
| 2 | 49.23 | 3.90 | $x_2 = 1.6922$ | $y_2 = 0.5910$ | 1.000 | |
| 3 | 48.67 | 4.77 | $x_3 = 1.6872$ | $y_3 = 0.6785$ | 1.1447 | |
| 4 | 45.76 | 9.87 | $x_4 = 1.6604$ | $y_4 = 0.9943$ | 1.6509 | |
| 5 | 42.58 | 28.27 | $x_5 = 1.6260$ | $y_5 = 1.4502$ | 2.3580 | |

$\Sigma x = 8.36 \quad \Sigma y = 4.18 \quad \Sigma xy = 6.954 \quad \Sigma x^2 = 14$

$y = ax + b$
$y_1 = ax_1 + b$
$y_2 = ax_2 + b$
$y_3 = ax_3 + b$
$y_4 = ax_4 + b$
$y_5 = ax_5 + b$

$\boxed{\Sigma y = a\Sigma x + b \cdot N}$

$y = ax + b$

$\Sigma y = a\Sigma x + b \times 5$

$4.18 = a \times 8.36 + b \times 5 \longrightarrow (1)$

$xy = ax^2 + bx$ | Multiplying by x both side

$\Sigma xy = a\Sigma x^2 + b \times \Sigma x$

$6.954 = a \times 14 + b \times 8.36 \longrightarrow (2)$

By eq$^n$ (1) + eq$^n$ (2)

$a = -$ ~~9099~~ $= 14.28$

$b = 24.71$

$y = ax + b$

$y = -14.28x + 24.71$

$VT^n = C$

$\log V + n \log T = \log C$

$x + ny = \log C$

$y = -\frac{1}{n}x + \frac{1}{n}\log C$

$\frac{1}{n} = 14.28$ $\qquad$ $\frac{1}{n}\log C = 24.71$

$n = \frac{1}{14.28} = 0.070$ $\qquad$ $\log C = 1.730$

$\underline{C = 54}$ Ans.

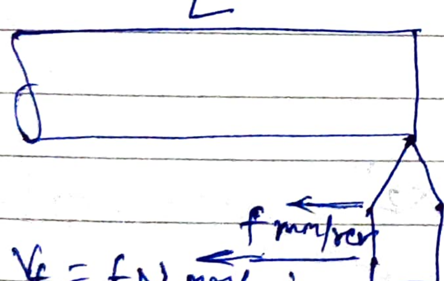
Gate 03

$V_1 = \pi D \times 50$ m/min

$V_2 = \pi D \times 80$ m/min

$V_3 = \pi D \times 60$ m/min

L



f mm/rev

$V_f = fN$ mm/min

$t = \frac{L}{fN}$ min.

10 Tools produce 500 comp
1 Tool — " — 50 com

$T_1 = 50 \times t_1 = 50 \times \frac{L}{f \times 50}$

$= \frac{L}{f}$ min.

$T_2 = 12.2 \times t_2 = 12.2 \times \frac{L}{f \times 80}$

$T_3 = x \times \frac{L}{f \times 60}$

$V_1 T_1^n = V_2 T_2^n = V_3 T_3^n$

$\cancel{\pi D} \times 50 \times \left(\frac{L}{f}\right)^n = (\cancel{\pi D} \times 80) \times \left(12.2 \frac{L}{f \times 80}\right)^n$

or $50 = 80 \times \left(\frac{12.2}{80}\right)^n$

or $\frac{5}{8} = \left(\frac{12.2}{80}\right)^n \Rightarrow n = 0.25$

$V_1 T_1^{0.25} = V_3 T_3^{0.25}$

$\cancel{\pi D} \times 50 \times \left(\frac{L}{f}\right)^{0.25} = \cancel{\pi D} \times 60 \times \left(\frac{x \times L}{f \times 60}\right)^{0.25}$

or $\frac{50}{6} = 60 \times \left(\frac{x}{60}\right)^{0.25}$

or $\frac{5}{6} = \left(\frac{x}{60}\right)^{0.25}$

or $x = 60 \times \left(\frac{5}{6}\right)^4$

$= 28.935 \simeq 29$

2 IES 06 conv. HSS Tool $T_1 = 1\,hr = 60\,min$

$V_1 = 30\,m/min$

$T_2' = 2.0\,min$

$V_2' = 60\,m/min$

$D = 300\,mm.$ for $T_3'' = 30\,min$

$V_1 T_1^n = C \; \cancel{V_2 T_2^n = V_3 T_3^n}$

$30 \times 60^n = C$

$60 \times 2^n = C$

$30 \times 60^n = 60 \times 2^n$

$\frac{60}{2} = 2^{1/n}$

$30 = 2^{1/n}$

$\log 30 = \frac{1}{n} \log 2$

$1.477 = \frac{1}{n} \times 0.3010$

$n = 0.2038$

$V_1 T_1^n = V_3 T_3^n$

$30 \times 60^{0.2038} = V_3 \times (30)^{0.2038}$

$V_3 = 35$ m/min

$V_3 = \pi DN \Rightarrow N = \frac{35}{\pi \times 0.3} = 37.13$ rpm.

gate 09 — clearance angle changed from 10° to 7° ($\alpha$).

[Hint: flank wear proportion to $\cot\alpha$

rake angle zero  tool life = ?

Tool life $\propto \frac{1}{\text{Tool wear}}$

$T \propto \frac{1}{\cot\alpha}$

$T \propto \tan\alpha$

$T = k\tan\alpha$

$T_1 = k\tan 10$

$T_2 = k\tan 7$

$\frac{T_2 - T_1}{T_1} \times 100\% = \frac{k\tan 7 - k\tan 10}{\tan 10} \times 100$

$= -30\%$

approx change in tool wear 30% decrease.

09 IES conventional

$T_0 = \left(T_c + \frac{C_t}{C_m}\right)\left(\frac{1-n}{n}\right)$

$T_c = 3$ min  $C_t = 3\text{min} \times 0.5 + 5 =$ Rs. 6.5/grind

$C_m = 0.50$ per min

$n = 0.2$

$T_0 = \left(3 + \frac{6.5}{0.50}\right)\left(\frac{1-0.2}{0.2}\right) = 64$ min

$V_0 = \frac{C}{T_0^n} = \frac{60}{64^{0.2}} = 26$ m/min

ESE 01 conv.

$V_1 = 50$ m/min $\quad T_1 = 45$ min

$V_2 = 100$ m/min $\quad T_2 = 10$ min

$V_1 T_1^n = C$

$V_2 T_2^n = C$

$V_1 T_1^n = V_2 T_2^n$

$50 \times 45^n = 100 \times 10^n$

$2 = (4.5)^n$

$\log 2 = n \log 4.5$

$n = 0.46$

$50 \times 45^{0.46} = C$

$C = 288$

Max^n productivity for 2 mins $\quad V_0 = ?$

$T_0 = T_c \left(\frac{1-n}{n}\right)$

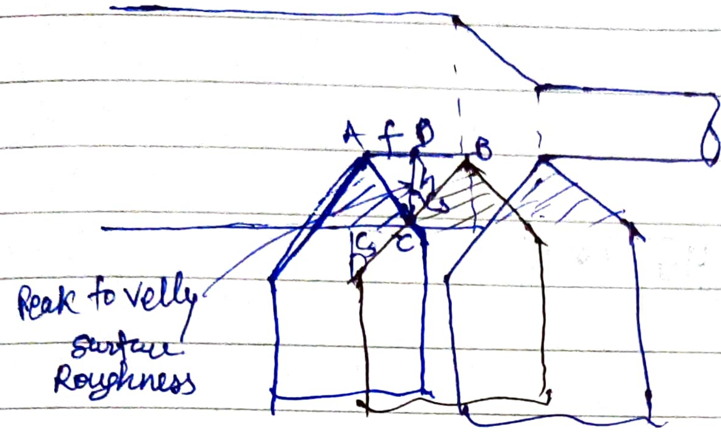
$= 2\left(\frac{1-0.46}{0.46}\right)$

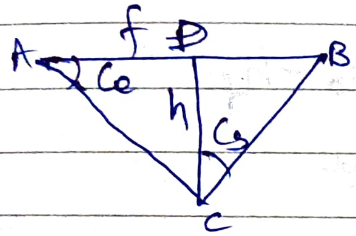
$= 2.34$ min.

$V_0 = \frac{C}{T_0^n} = \frac{288}{2.34^{0.46}} = 194$ m/min

# Turning

## Surface Roughness (In Turning Operation):




$$\frac{AD}{h} = \cot C_e$$

$$\frac{DB}{h} = \tan C_s$$

$$AD + BD = f$$

$$h \cot C_e + h \tan C_s = f$$

Roughness. $$\boxed{h = \frac{f}{\tan C_s + \cot C_e}}$$

Ideal surface [zero nose Radius.

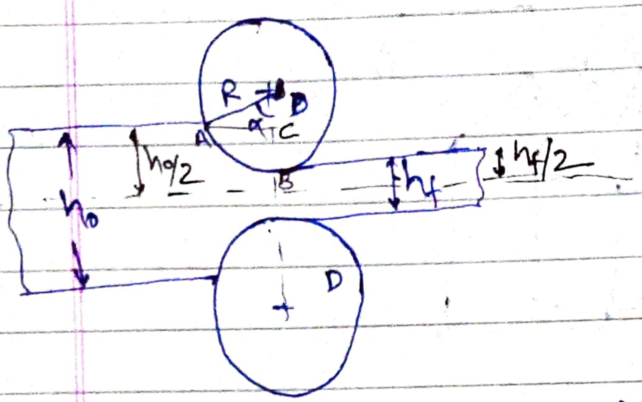
$R_a$ or CLA (center line average) $= \frac{h}{4}$

↓ Roughness average

with nose Radius $\boxed{h = \frac{f^2}{8R}}$ & $\boxed{R_a = \frac{f^2}{18\sqrt{3}R}}$

# Rolling

## Formula in Rolling:-




Reduction or Draft $(\Delta h) = h_0 - h_f$

Arc of contact $(AB) = R \cdot \alpha$ ← Rad.

$\alpha \rightarrow$ angle of bite

$$BC = \frac{h_0}{2} - \frac{h_f}{2} = \frac{\Delta h}{2}$$

$$OC = R - \frac{\Delta h}{2}$$

$$\cos\alpha = \frac{OC}{OA} = \frac{R - \Delta h/2}{R}$$

or $R\cos\alpha = R - \frac{\Delta h}{2}$

or $\frac{\Delta h}{2} = R - R\cos\alpha$

or $\boxed{\Delta h = 2R(1-\cos\alpha)}$

or $\boxed{\Delta h = D(1-\cos\alpha)}$ VIMP ***

Gate 07

$h_0 = 16$ mm $h_f = 10$ mm $D = 400$ mm

$\Delta h = h_0 - h_f = D(1-\cos\alpha)$

$16 - 10 = 400(1-\cos\alpha)$

$\cos\alpha = 1 - \frac{6}{400} = \frac{394}{400}$

$\underline{\alpha = 9.936^\circ \; \cancel{rad}.}$

Gate 12

$D = 410$ mm $h_0 = 140$ mm $\Delta h_f = 8$ mm Reduction $h_f$ 10%.

$\Delta h = 0.8$

$0.8 = 410(1-\cos\alpha)$

$\cos\alpha = 1 - \frac{0.8}{410}$

$\alpha = 3.579^\circ = \frac{3.579 \times \pi}{180}$ rad.

<u>Unaided entry</u> (Rolling).

$\mu \geq \tan\alpha$.

$\mu = \tan\alpha$ [extreme condn]

Maxm Reduction or Maxm draft. $\boxed{\Delta h_{max} = \mu^2 R}$.

$h_0 - h_{f\,min} = \mu^2 R$.

$\boxed{h_{f\,min} = h_0 - \mu^2 R}$

Minm possible thickness in a single pass.

No. of pass required $N = \frac{(\Delta h)_{required}}{(\Delta h)_{max}}$.

Gate

$h_o = 4mm$ $D = 300 mm$

$\mu = 0.1$

$h_{f\,min} = h_o - \mu^2 R$

$= 4 - (0.1)^2 \times 150$

$= 2.5$ mm.

Gate''

$h_o = 30 mm$ $h_f = 10 mm$

$D = 600 mm$ $\mu = 0.1$

$N = \dfrac{\Delta h_{require}}{\Delta h_{max}}$

$\Delta h = 30 - 10 = 20 mm$

$\Delta h_{max} = \mu^2 R = (0.1)^2 \times 300$

$= 3$

$N = \dfrac{20}{3} = 6.66 \simeq 7$

IES'01

$h_o = 30 m$ $h_f = 15 m$ $D = 300$

$\mu \geq \tan\alpha$. $\mu = ?$

$\Delta h = h_o - h_f = D(1 - \cos\alpha)$

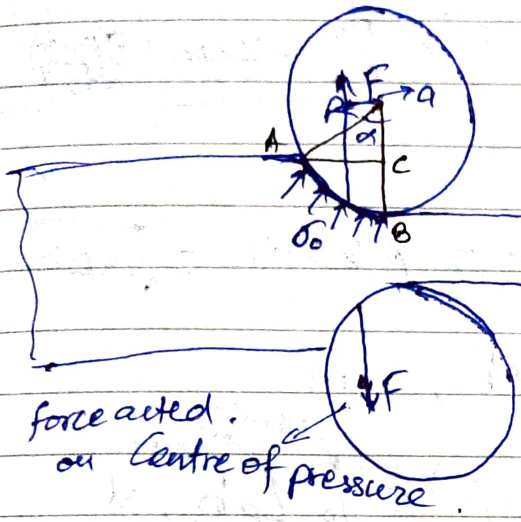
$\cancel{\Rightarrow 15 = 300}$

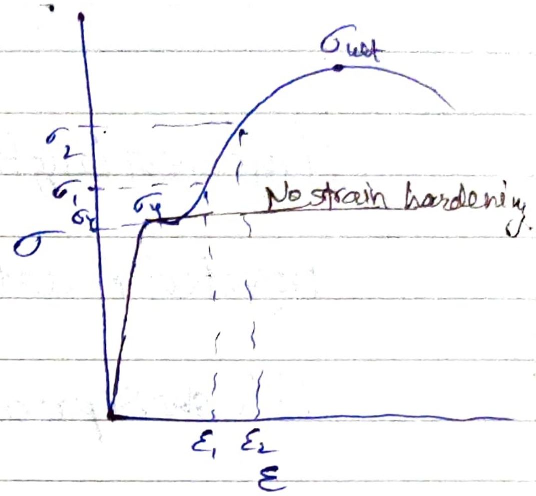
$\Delta h_{max} = \mu^2 R$

$30 - 15 = \mu^2 \times 150$

$\mu^2 = \sqrt{\dfrac{15}{150}} = \sqrt{\dfrac{1}{10}} = \dfrac{1}{3} = 0.33$

# Force, Torque & Power in Rolling :-



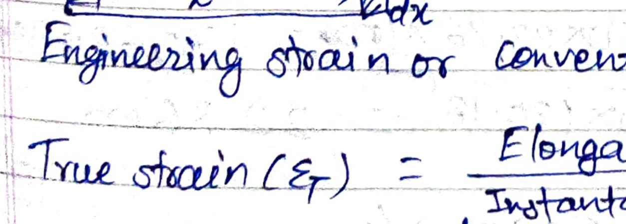

$\sigma_0$ = flow stress.

$= K\varepsilon_T^n$ with strain hardening

$\sigma_0 = \sigma_y$ without strain hardening.

True strain ($\varepsilon_T$)

$$\varepsilon_T = \ln(1+\varepsilon) = \ln\frac{L}{L_0}$$




Engineering strain or conventional strain $\varepsilon = \frac{\Delta L}{L_0}$

True strain ($\varepsilon_T$) $= \frac{\text{Elongation}}{\text{Instantaneous length}}$

$$= \int_{L_0}^{L} \frac{dx}{x} = \ln\frac{L}{L_0}$$

$$= \ln(1+\varepsilon)$$

$$\frac{L}{L_0} = \frac{L_0+\Delta L}{L_0} = 1+\frac{\Delta L}{L_0} = 1+\varepsilon$$

[In forging negative value]

$$A_0 L_0 = AL \text{ or } \frac{L}{L_0} = \frac{A_0}{A} = \frac{d_0^2}{d^2}$$

$$\boxed{\varepsilon_T = \ln(1+\varepsilon) = \ln\frac{L}{L_0} = \ln\frac{A_0}{A} = 2\ln\left(\frac{d_0}{d}\right)}$$

Projected length $(L_p) = AC = R\sin\alpha = \sqrt{R\Delta h}$ mm

Projected Area $(A_p) = L_p \times b$, $mm^2$

Roll ~~sho~~ Separating Force $(F) = \underset{N/mm^2}{\sigma_0} \underset{mm}{L_p} \underset{mm}{b}$, N

Arm length $(a) = 0.5 L_p$ for hot rolling
$= 0.45 L_p$ for cold rolling } mm

Torque per roller $(T) = \frac{\underset{N}{F} \times \underset{mm}{a}}{1000}$, N.m

Total Power for two rollers $(P) = \underline{2 T \cdot \omega}$, W.

Ques Gate 08.

$h_0 = 20$ mm  $b = 100$ mm  $h_f = 18$  $R = 250$ mm
$N = 10$ rpm  $\sigma_0 = 300$ MPa  $P = ?$

$F = \sigma_0 L_p b = 300 \times 22.36 \times 100 = 670{,}820.39$ N

$L_p = \sqrt{R\Delta h} = \sqrt{250 \times (20-18)} = 22.36$ mm

$P = 2 \cdot T \cdot \omega = 2 \times F \times a \times \omega / \cancel{1000}$  $\boxed{\omega = 2\pi N}$

$= \frac{2 \times 670820.39 \times 0.45 \times 2\pi \times 10 \times 22.36}{1000 \times 1000 \times 60}$ kW

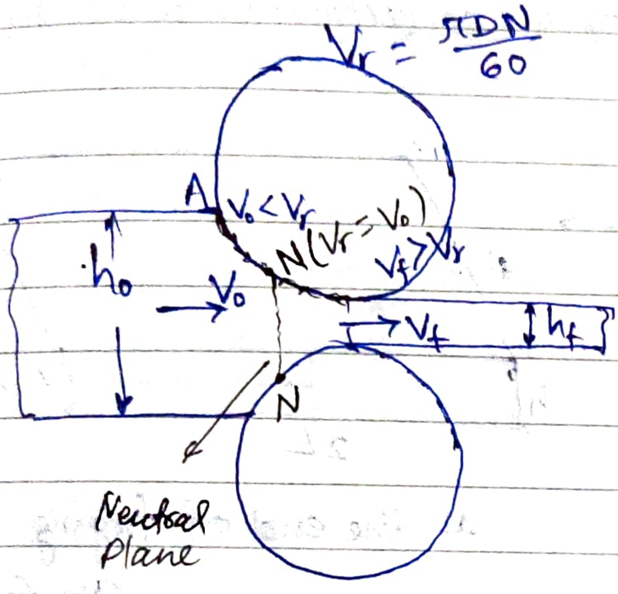
$\approx 15.69$ kW

Teacher's Signature

Note

In rolling, force × Velocity = Power, formula is not applicable becoz dir^n of force & dir^n of velocity are not same.

## Neutral Plane:-




$$\text{Backward slip} = \frac{V_r - V_o}{V_r} \times 100\%$$

$$\text{Forward slip} = \frac{V_f - V_r}{V_r} \times 100\%$$

Gate 07 $D_1 = 200\text{mm}$ $D_2 = 400\text{mm}$ $h = 60\text{mm}$.

TB $\varepsilon_T = ?$

IES 08

Ground → Grinding

Hot forged on hammer

Heat treatment.

Polishing.